

THE ADVERSARIAL REASONING METHOD

A Conceptual Framework Proposal for Structured Clinical Diagnostic Reasoning

Not Yet Empirically Validated

Proposed for Pilot Evaluation

Working Draft & White Paper

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July 2026

Abstract

Diagnostic error remains among the most widespread and consequential sources of preventable harm in modern medicine. A substantial share of these errors reflects failures of the reasoning process rather than just gaps in medical knowledge (premature closure, confirmation bias, anchoring, and the absence of deliberate self-critique).

This paper proposes the Adversarial Reasoning Method (ARM), structured by an eight-step framework for clinical diagnostic reasoning organized around the steps as listed: Observe, Represent, Compete, Attack, Challenge, Update, Reflect, and Integrate. Each individual step targets a specific, previously documented failure mode in clinical cognition.

The framework is presented explicitly and only as a conceptual proposal. It has not been empirically tested, and no claim of superiority over existing models of clinical reasoning is made. Where the framework's components do overlap with prior published work, most directly the problem-representation model of Bordage, hypothetico-deductive method of Kassirer, and the cognitive forcing strategies of Croskerry, this paper says so plainly and identifies exactly what, if anything, is added. Where no defensible claim to originality exists, this paper describes the contribution as a synthesis rather than an invention.

This paper also proposes a small-scale case-series pilot as a feasibility step preceding any larger validation study and addresses two practical concerns that are often absent from clinical-reasoning frameworks: the conflict between procedural rigor and clinical throughput, and the operational constraints of introducing structured adversarial documentation into real clinical/educational settings.

(The intended audience for this document is clinician educators, program directors, and medical-education researchers who might sponsor, critique, or help design a pilot evaluation. It is not intended as a finished curriculum, a validated instrument, or a claim of practice-ready status.)

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1. The Problem: Why Diagnostic Errors Occur

1.1 Scope

Diagnosis is among the most fundamental responsibilities in medicine. Diagnostic errors can lead to delayed treatment, inappropriate intervention, avoidable cost, and patient harm.

In the United States, approximately 12 million adults experience a diagnostic error in the outpatient setting annually (Singh et al., 2014). Autopsy studies have consistently demonstrated a 10 to 40 percent discordance rate between clinical and post-mortem diagnoses, a finding that has remained largely stable for decades despite advances in imaging, laboratory medicine, and information technology (Hoppe et al., 2025). The National Academy of Medicine has identified improving diagnosis as a critical health-system priority (National Academies of Sciences, Engineering, and Medicine, 2015).

1.2 Cognitive Sources of Diagnostic Error

Diagnostic reasoning involves both intuitive pattern recognition and analytical evaluation. Errors often arise when rapid, intuitive judgements are not properly checked against new or disconfirming information. The following biases are among the most frequently implicated:

- Anchoring bias: excessive commitment to the initial diagnostic impression
- Premature closure: accepting a diagnosis before sufficient evidence has been gathered
- Confirmation bias: selectively seeking evidence that supports an existing hypothesis
- Availability bias: overestimating the probability of vivid or recently encountered diagnoses
- Framing effects: allowing referral source or case narrative to constrain the diagnostic search
- Base-rate neglect: failing to weight prior probability appropriately

Many diagnostic errors occur despite adequate medical knowledge and are better understood as failures of the reasoning process itself.

1.3 Reasoning Under Uncertainty

Clinicians routinely act on incomplete histories, evolving presentations, and imperfect tests. Effective diagnostic reasoning depends on managing that uncertainty without converging prematurely on a single explanation.

1.4 The System-Level Contribution

Cognitive factors do not fully explain diagnostic error. Time constraints, throughput pressure, EHR burden, weak diagnostic feedback loops, and hierarchical communication failures compound individual cognitive vulnerabilities. Any framework intended for real use must remain usable under these constraints, not only in idealized teaching settings — a point this paper returns to directly in Sections 9 and 10.

1.5 Problem Statement

Existing educational approaches provide strong instruction in clinical knowledge and differential diagnosis, but fewer frameworks explicitly walk a clinician through structured hypothesis generation,

self-critique, uncertainty monitoring, and post-case reflection as a single connected sequence. This paper proposes such a sequence and is explicit about which parts of it are new and which are not.

2. Existing Approaches and Their Limitations

2.1 Illness Script Theory

Illness scripts encode the enabling conditions, mechanisms, and characteristic findings of a disease. Expert clinicians build rich illness scripts through repeated exposure, which enables fast, often accurate pattern matching. The theory is primarily descriptive: it explains expert retrospective reasoning, but offers no normative procedure for novices, atypical presentations, or systematic bias detection.

2.2 Hypothetico-Deductive Reasoning

Kassirer's hypothetico-deductive model describes generation of a small set of candidate diagnoses early in the encounter, followed by targeted information gathering to confirm or refute each one. It has strong empirical support across levels of expertise. Its main limitation is the absence of a formal mechanism for detecting when a hypothesis has been inadequately challenged — clinicians frequently gather confirming evidence without systematically seeking disconfirming data, an error the model does not structurally prevent.

2.3 Differential Diagnosis Approaches

Traditional differential diagnosis generates and progressively narrows multiple candidate explanations. It guards against fixation on a single diagnosis but offers limited guidance on how competing diagnoses should be challenged, prioritized, or retrospectively evaluated after resolution.

2.4 Bayesian Reasoning

Bayesian approaches formalize updating of diagnostic probability using prior probability and likelihood ratios. They provide a mathematically rigorous account of belief revision but formal real-time computation is rarely feasible: clinicians lack precise likelihood ratios for most findings, priors are highly context-dependent, and tracking more than a few hypotheses numerically exceeds practical cognitive capacity. In most clinical settings, Bayesian reasoning functions as a conceptual orientation, not a real-time calculation.

2.5 Dual-Process Theory Interventions

Educational interventions built on dual-process theory, including Croskerry's cognitive forcing strategies, prompt clinicians to pause and apply deliberate analysis before finalizing a diagnosis. Evidence for sustained real-world effectiveness is mixed. The core limitation is structural: metacognitive awareness without a concrete procedural scaffold produces inconsistent results. Clinicians may recognize the concept of bias without a repeatable method for identifying and correcting it mid-reasoning.

2.6 Why These Approaches Remain Complementary Rather Than Competitive

ARM is not proposed as a replacement for illness scripts, Bayesian reasoning, differential diagnosis or hypothetico-deductive reasoning. It is proposed as a procedural sequence within which these approaches can operate: illness scripts inform hypothesis generation, Bayesian principles can inform probability revision, and differential diagnosis defines the competing possibilities under consideration. Section 3 makes this relationship explicit.

3. Positioning Relative to Bordage, Kassirer, and Croskerry

Readers familiar with the clinical reasoning literature will recognize substantial overlap between ARM's steps and three specific published models: Bordage's work on problem representation and semantic qualifiers, Kassirer's hypothetico-deductive method, and Croskerry's cognitive forcing strategies. This section states that overlap directly rather than leaving it to be discovered, and separates what is adopted from what is claimed as new.

3.1 Bordage: Problem Representation

Bordage's account of problem representation — compressing a case into a compact, semantically qualified summary (for example, framing findings by tempo, severity, and pattern rather than a raw symptom list) — is the direct source of ARM's Step 2 (Represent). ARM adds no new theoretical content here. It adopts Bordage's model and mandates it as a required checkpoint before hypothesis generation begins, rather than leaving problem representation as an informal or optional habit.

3.2 Kassirer: Hypothetico-Deductive Reasoning

Kassirer's model of early hypothesis generation followed by targeted, confirmatory-or-refuting data gathering underlies ARM's Steps 1, 3, 5, and 6 (Observe, Compete, Challenge, Update) taken together. ARM does not introduce a new cognitive mechanism here. What differs is procedural: Kassirer's model does not itself mandate a distinct adversarial step against the leading hypothesis, and does not specify a formal post-hoc reflection or long-term integration stage. ARM's contribution in this region is sequencing and mandatory documentation, not new theory.

3.3 Croskerry: Cognitive Forcing Strategies

Croskerry's "consider the opposite" strategy and broader catalogue of cognitive forcing strategies are the direct antecedent of ARM's Step 4 (Attack). Croskerry presents disconfirmation-seeking as one situational tool among several bias-mitigation strategies, to be deployed at the clinician's discretion. ARM's difference is that it makes this step non-optional and places it at a fixed point in a whole-case sequence, paired with a structured set of forced questions (what evidence would exist if I am correct; if I am wrong; what would a disagreeing colleague argue) rather than a general reminder to consider alternatives.

3.4 Summarized Comparison

Feature	Bordage	Kassirer	Croskerry
Structured problem representation	Originates it	Not specified	Not specified
Early hypothesis generation & testing	Not specified	Originates it	Partial (situational)
Disconfirmation as mandatory step	No	No	Proposes as optional tool
Formal post-case reflection stage	No	No	No
Long-term pattern integration stage	No	No	No

3.5 Honest Statement of Contribution

Taken together, ARM's original contribution is limited to three things: sequencing previously separate patterns into one ordered procedure; converting Croskerry's optional debiasing tool into a mandatory, fixed-position step; and closing the loop with explicit Reflect and Integrate stages, which have no clear precedent across these three models. Steps 1, 2, 3, 5, and 6, are, in substance, procedural restatements of Bordage's and Kassirer's published work. Where this paper cannot identify a defensible claim to novel cognitive content, it says so here rather than implying otherwise: ARM is best described as *a synthesis and procedural sequencing of existing, separately published models*, with the Attack step's mandatory placement and the Reflect/Integrate stages as its only structurally original elements.

4. The Adversarial Reasoning Method (ARM): Framework Overview

4.1 Design Principles

ARM rests on four premises:

- Many diagnostic errors reflect failures of reasoning process rather than deficiencies in medical knowledge alone.
- Hypothesis testing requires deliberate adversarial effort; disproving one's own leading diagnosis is a different cognitive act than passively accumulating supporting evidence.
- Clinical intuition is compressed expertise, not guesswork, and should be trained and interrogated rather than suppressed or blindly trusted.
- Diagnostic certainty is provisional; effective reasoning requires continuous reassessment of confidence and willingness to revise favored explanations.

The framework is intended to be used stepwise in high stakes or undifferentiated cases and selectively compressed in routine ones (see Section 9). It is a cognitive scaffold, not a mandatory protocol for every encounter.

4.2 The Eight Steps at a Glance

#	Step	Core Function	Cognitive Failure Addressed
1	Observe	Structured intake of clinical/contextual data; separation of observation from interpretation	Incomplete data gathering
2	Represent	Compression of findings into a concise, pattern-oriented summary	Diagnostic ambiguity
3	Compete	Generation of competing hypotheses with supporting/contradicting evidence	Premature closure
4	Attack	Adversarial audit of the leading hypothesis; explicit bias identification	Confirmation bias, anchoring
5	Challenge	Test selection based on capacity to alter the differential	Low-value testing
6	Update	Iterative revision as new data accumulate	Diagnostic inertia
7	Reflect	Structured post-case self-evaluation	Failure to learn
8	Integrate	Long-term pattern formation across cases	Passive experience accumulation

Although presented sequentially for instructional clarity, movement between steps is iterative. New information may prompt renewed observation, reformulated problem representation, or repeated adversarial testing.

5. Step-by-Step Explanation

Step 1: Observe: Case Intake and Symptom Inventory

- Document age, sex, occupation, environmental exposures, symptom timeline, past medical history, medications, allergies, lifestyle factors, and psychosocial context.
- Separate positive findings from negatives; absence is often as diagnostically significant as presence.
- Distinguish observation from interpretation: “patient reports chest pain” is data; “cardiac chest pain” is a hypothesis. These must not be conflated at this stage.

Step 2: Represent: Problem Representation

- Use age, tempo, and system involvement as the primary framing elements (Bordage).
- Transform a symptom list into a diagnostic pattern, e.g. “subacute multisystem inflammatory illness in a previously healthy middle-aged man.”
- Avoid premature diagnostic language; “multisystem inflammatory process,” not “lupus.”

Step 3: Compete: Hypothesis Generation

- For each hypothesis, document supporting evidence, contradicting evidence, relative likelihood, and consequences if missed.
- Document intuitive impressions separately from formal reasoning; they are a distinct cognitive signal worth interrogating, not a substitute for evidence.
- Likelihood is expressed comparatively by default (“more likely than,” roughly equally likely”). See section 6 for the rationale and an optional numerical extension.
- Hypothesis count should scale with uncertainty: one or two for a clear presentation, five or more for an undifferentiated one.

Step 4: Attack: Bias and Assumption Audit

- Name the biases that may be operating: anchoring, availability, confirmation, framing, base-rate neglect.
- For the leading hypothesis, ask: What evidence must exist if I am correct? If I am incorrect? What single finding would most challenge my model? What am I most likely overlooking? What would a disagreeing colleague argue?
- Identify the single best alternative hypothesis that accounts for the same data.

Step 5: Challenge: Test Selection and Evidence Mapping

- Prioritize tests capable of shifting probability across multiple competing diagnoses at once, not only confirming the leading one.
- Weigh sensitivity, specificity, false-positive/negative rates, invasiveness, risk, and cost against the ability of the test to distinguish among hypotheses.

- Document the expected impact of each possible result on the differential before the result is known.

Step 6: Update: Iterative Revision

- Classify prior assumptions as correct, incorrect, or irrelevant in light of new data.
- Actively reconsider previously dismissed diagnoses — new data may rehabilitate a low-probability hypothesis.
- Avoid interpreting ambiguous new data as automatically confirming the existing impression; confidence should move up, down, or stay level based on the actual quality of new evidence.

Step 7: Reflect: Post-Case Analysis

- Ask: What did I observe correctly? What did I miss? Which biases influenced my reasoning? What changed my model, and when?
- Identify near-misses: diagnoses almost missed, held too long, or investigated too late.
- Reflect regardless of outcome — a correct diagnosis can still result from flawed reasoning that will eventually fail

Step 8: Integrate: Pattern Formation and Long-Term Learning

- Compare cases over time to identify presentation patterns and recurring cognitive vulnerabilities.
- Distinguish genuine pattern recognition from narrative coherence; a case “making sense” in hindsight does not validate the process that reached the conclusion.
- Intuition developed through systematic reflection may be more reliable than intuition from passive experience alone.

6. Optional Numerical Extension for Advanced Use

6.1 Why Comparative Language Is the Default

Steps 3 and 6 use comparative likelihood language (“more likely than.” “much less likely unless”) rather than numerical probability estimates by default, for three specific reasons rather than by omission:

- Cognitive load: assigning and tracking numerical priors and posteriors across several competing hypotheses in real time exceeds practical working-memory capacity for most clinicians, particularly under time pressure (see Section 2.4).
- Calibration error risk: untrained clinicians asked to produce numbers tend to produce false precision: a percentage that feels rigorous but is not actually derived from a likelihood ratio or a real prior. Overconfidence built on invented numbers is arguably worse than acknowledged qualitative uncertainty (Berner & Graber, 2008).
- Portability: comparative language transfers across specialties and experience levels without requiring shared numerical literacy or agreed-upon likelihood ratios, which rarely exist for most clinical findings.

6.2 Optional Extension for Trained or Research Users

For clinicians or research contexts with genuine training in Bayesian reasoning, an optional numerical layer can be applied at Steps 3 and 6 without changing the framework’s structure:

- At Step 3, assign a rough prior probability band to each hypothesis (e.g., <10%, 10-50%, >50%) rather than a false-precision point estimate, or a full prior where a defensible base rate exists.
- At Step 6, apply likelihood ratios where they are actually known (e.g., a well-characterized laboratory test) to compute a posterior; otherwise retain the qualitative default rather than creating a likelihood ratio.

This extension is opt-in and is not recommended as a default for trainees without prior instruction in Bayesian updating, for the calibration reasons stated in 6.1. A pilot evaluating ARM (Section 8) could usefully test whether the numerical extension improves or degrades reasoning quality relative to the comparative default among users with and without prior Bayesian training.

7. Worked Case

The following case demonstrates application of all eight steps to a common, instructive presentation. This is a single illustrative example, not evidence of the framework's effectiveness; Section 8 proposes how a small series of such cases could be used as a pilot.

Step	Application
1: Observe	22-year-old woman. Fatigue extending 3 months, pallor, palpitations on exertion, brittle nails, dysphagia. Menorrhagia. No medications. Vegetarian diet (limited red meat). RBC low, MCV low, serum ferritin 4ng/mL. Negative factors: no weight loss, no lymphadenopathy, no bleeding history beyond menstrual.
2: Represent	Microcytic, hypochromic anemia with an iron-deficiency pattern in a young woman with combined dietary and menstrual iron loss.
3: Compete	Leading: iron deficiency anemia (IDA). Competing: thalassemia trait (microcytic but ferritin typically normal); anemia of chronic disease (ferritin typically elevated). Rare: sideroblastic anemia. IDA supported by ferritin, diet, menses; thalassemia retained pending hemoglobin electrophoresis.
4: Attack	Is IDA too obvious a fit; confirmation bias risk? Test: if IDA, oral iron should produce a reticulocyte response in 7-10 days; if not, reconsider. Could IDA and thalassemia trait coexist?
5: Challenge	CBC with indices; ferritin, TIBC, transferrin saturation; hemoglobin electrophoresis to exclude thalassemia; peripheral smear. If no response, consider a GI source (endoscopy).
6: Update	Ferritin of 4 confirms IDA. Electrophoresis normal; thalassemia excluded. Reticulocyte surge at day 9 confirms diagnosis. Revised differential: IDA as sole diagnosis.
7: Reflect	Availability bias was a real risk — IDA is common in young women. The value of electrophoresis was almost skipped. Explicit hypothesis competition prevented premature closure.
8: Integrate	Pattern filed: microcytic anemia + low ferritin + young female + dietary restriction = high IDA pretest probability; perform electrophoresis in populations with meaningful thalassemia prevalence; reticulocyte response is the confirmatory test of record.

Teaching point: the value in this case was not the diagnosis. IDA is readily identified. The value was the explicit retention of a competing hypothesis, the disconfirming test that excluded it, and the confirmatory step that moved the diagnosis from plausible to confirmed. The framework forces documentation of reasoning rather than conclusions.

8. Proposed Case-Series Pilot

A single worked case, however clean, demonstrates only that the framework can be applied, not that it improves reasoning or is usable as scale. Before any randomized or simulation-cohort trial (Section 12), a small case-series pilot is proposed as a feasibility step.

8.1 Purpose

The pilot's purpose is feasibility and instrumentation, not efficacy. It is designed to answer operational questions that a larger trial would otherwise have to guess at:

- Do independent users of the framework converge on similar Attack-step outputs (leading hypothesis, best alternative, key disconfirming question) for the same case?
- How long does full documentation actually take, per step, across case types of varying complexity?
- Which steps compress naturally without loss, and which are skipped under time pressure even when the case warrants them?
- Does the framework surface a near-miss or reconsidered diagnosis that unstructured reasoning on the same case would plausibly have missed?

8.2 Proposed Design

- Assemble 5-8 de-identified case vignettes spanning a range of acuity, specialty, and diagnostic difficulty (include at least one straightforward case and at least one genuinely ambiguous one).
- Recruit 2-4 raters of comparable training level (e.g., senior medical students or resident-level reviewers) to independently apply the full eight-step ARM documentation to each case.
- Time each step per rater per case.
- Score inter-rater agreement on: (a) identity of the leading hypothesis, (b) identity of the named best alternative, (c) whether a disconfirming test was proposed at all, and (d) whether a near-miss or previously dismissed diagnosis was reconsidered at Step 6.
- Collect brief qualitative feedback from raters on which steps felt redundant, unclear, or artificially forced.

8.3 What This Pilot Would and Would Not Show

A favorable result — reasonable agreement on best-alternative identification, manageable completion times, and raters reporting the Attack step changed their confidence — would justify investment in the larger designs described in Section 12. It would not, on its own, demonstrate reduced diagnostic error, since case vignettes are not real patients and small convenience samples of raters are not powered for anything beyond feasibility. This distinction is stated explicitly so the pilot is not later mischaracterized as validation.

9. Rigor Versus Throughput

The full eight-step sequence is deliberately System-2-heavy: it asks for explicit documentation at each stage, which takes real time. Real clinical environments are throughput-constrained, and a framework that ignores that conflict is not usable, whatever its theoretical merit.

Rather than treating this as an afterthought, a tiered application is proposed:

Tier	Trigger	Recommended Depth
Tier 1: Routine	Clear presentation, high pretest probability, low stakes if wrong.	Observe > Represent only, applied informally without documentation
Tier 2: Moderate	Some diagnostic uncertainty or moderate consequence if wrong	Observe > Represent > Compete > Attack (the “minimum viable ARM”), briefly documented
Tier 3: High Stakes / Undifferentiated	Diagnostic uncertainty is high, consequences of error are serious, or presentation is atypical	Full eight-step sequence with documentation, including Reflect and Integrate after resolution

No data currently support these specific tier thresholds or the boundary between them; they are a proposed heuristic, not an established one. The case-series pilot in Section 8 could reasonably be extended to test whether raters correctly and consistently classify cases into these tiers before recommending the thresholds for adoption.

10. Implementation Constraints

Beyond time cost, several practical constraints would need to be resolved before any pilot moves from paper to a real educational or clinical setting:

- Documentation burden and EHR integration: an ARM worksheet added on top of existing note-writing requirements competes for the same limited time as clinical documentation itself; any real deployment likely needs a lightweight, separate template rather than an addition to the permanent chart note.
- Medico-legal exposure: written adversarial reasoning (explicitly documenting “what would prove me wrong” or a dismissed alternative diagnosis) could be viewed differently in a discoverable medical record than in an educational worksheet. A prudent implementation keeps ARM documentation in an educational or quality-improvement context (e.g., a training exercise or M&M conference tool) rather than the permanent chart, at least until institutional and legal stakeholders have weighed in.
- Institutional and cultural buy-in: attendings and programs unfamiliar with mandatory self-disconfirmation exercises may perceive them as inviting some sort of scrutiny of uncertainty rather than improving it; a pilot needs at least one faculty champion willing to model the Attack and Reflect steps openly.
- Training burden: the Attack and Reflect steps require intellectual honesty and tolerance for being shown wrong in front of peers or supervisors; without some baseline psychological safety, learners are likely to perform the steps superficially rather than genuinely.
- Scalability of rater/grader time: any pilot or curricular use that requires faculty review of ARM worksheets adds grading burden on top of existing teaching loads, which needs to be budgeted for explicitly rather than assumed to be free.

11. Strengths and Limitations

11.1 Strengths

- Procedural completeness: the eight steps span the full arc of a clinical encounter, from intake to long-term learning, in one integrated sequence.
- Mandatory adversarial step: making Croskerry’s “consider the opposite” strategy as a fixed, non-optional step is a small but concrete procedural difference from treating it as one situational tool among many.
- Named bias specificity: rather than a generic warning about cognitive bias, the framework pairs the attack step with specific biases and specific interrogating questions.
- Compatibility with existing models: the framework is explicit that it organizes, rather than replaces, illness scripts, Bayesian updating, and hypothetico-deductive reasoning (Section 3)
- Education tractability: each step has a defined purpose and a documentable output, which lends itself to teaching and to the pilot design in Section 8.

11.2 Limitations

- No empirical validation: no prospective, simulation-based, or randomized data currently exist showing that ARM reduces diagnostic error. This is a structural gap, not a nuance, and this paper makes no claim to the contrary.
- Full completion of all eight steps adds real overhead; Section 9 proposes a tiering scheme to address this, but the thresholds are untested.
- Quantification of uncertainty: The default framework is qualitative. Section 6 addresses this directly.
- Implementation friction: Documentation burden, medico-legal exposure, and institutional buy-in (Section 10) are real barriers to adoption beyond a controlled pilot setting.
- Learning curve: The Attack and Reflect steps require tolerance for uncertainty, which may take training and a supportive environment to properly develop.

12. Future Development

12.1 Sequencing

This paper proposes the following sequence, each stage gated on the success of the one before it:

- Stage 1 (this paper): conceptual specification and internal consistency review.
- Stage 2: the case-series feasibility pilot described in Section 8.
- Stage 3: a simulation-based cohort study, in which medical students or residents apply ARM to standardized vignettes with pre-post assessment of diagnostic accuracy and bias-identification scores.
- Stage 4: a randomized controlled trial comparing diagnostic accuracy and error rates with a primary outcome of diagnostic error reduction as adjudicated by blinded expert panel review.

Moving to Stage 3 and/or 4 without a successful Stage 2 would mean testing an instrument whose basic usability and inter-rater consistency have not been established, which is a design flaw this roadmap is intended to prevent.

12.2 Integration With Medical Education (Longer-Term, Contingent on Validation)

- Pre-clinical: structured Compete/Attack drills in case-based learning, based on Stage 2/3 results.
- Clinical clerkships: a documentation tool for case presentations requiring explicit Represent and Attack steps.
- Residency: a structured Reflect/Integrate tool for morbidity and mortality conferences as an alternative to retrospective blame-oriented review.

12.3 Digital Integration (Speculative)

A structured ARM template embedded in an EHR or a standalone educational tool could, in principle, prompt documentation of competing hypotheses at the time of ordering and flag cases with no disconfirming test ordered against the leading hypothesis. This is listed as a long-term, speculative direction, not a near-term deliverable, and depends entirely on the medico-legal and workflow questions raised in Section 10 being resolved first.

12.4 Subspecialty Adaption (Speculative)

- Emergency medicine: a compressed Observe > Represent > Attack sequence suited to a high-acuity setting.
- Radiology: Attack/Update sequences adapted to image interpretation rather than clinical synthesis.
- Pathology/laboratory medicine: Compete/Attack logic applied to laboratory interpretation.

13. Conclusion

Diagnostic error is not an unsolvable problem, but it will not be solved by medical knowledge alone. The Adversarial Reasoning Method (ARM) proposes a structured, sequenced procedure for clinical diagnostic reasoning, built largely by organizing and making mandatory a small number of previously published ideas (Bordage's problem representation, Kassirer's hypothetico-deductive testing, and Croskerry's cognitive forcing strategies) and closing the loop with explicit reflection and pattern integration.

This paper makes no claim that ARM has been shown to reduce diagnostic error, and no claim of novelty beyond what Section 3 states plainly: sequencing, mandatory placement of the adversarial step, and the Reflect/Integrate stages. The framework is offered here as a conceptual proposal, together with a concrete, appropriately modest pilot design (Section 8) intended to generate the feasibility data needed before any larger claim could be responsibly made.

The goal of this document is not to present a completely original framework. The goal is a testable proposal, honestly scoped, that could plausibly contribute to fewer missed diagnoses if it survives the pilot stages outlined above, and an honest account of why it wouldn't.

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